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Design and Evaluation of a Legal AI Assistant: A Comparative Study of RAG and Fine-Tuning

Monthly Progress Report – June/July 2026

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Overview

This report covers the period from June 6 to July 3, 2026. It focuses on the fine-tuning pipeline, the retrieval component upgrade, and the comparative evaluation of the different system configurations.

Fine-Tuning Pipeline

First iteration recap

The first QLoRA fine-tuning run was completed at the end of May. Mistral-7B was adapted on 380 question-answer pairs built from BSARD, using a LoRA adapter with rank r=16 and 4-bit NF4 quantization to stay within the 16GB VRAM budget of the Colab T4 GPU. Training covered 3 epochs and took roughly 6 hours. The evaluation results on 50 test questions were as follows.

Run	ROUGE-L	BERTScore F1
First run (r=16, 380 examples)	0.1375	0.6814

Table 1: First fine-tuning run results on 50 test questions

The BERTScore of 0.68 indicates that the model produces answers that are semantically close to the reference articles even when the exact wording differs. The ROUGE-L is lower because the model tends to paraphrase rather than reproduce the text word for word, which this metric penalises even when the answer is factually correct.

Second iteration

Two modifications were made for the second run. The training set was extended from 380 to 580 examples by adding 200 pairs drawn from the synthetic question split of BSARD. These automatically generated questions were filtered manually to remove those that were ambiguous or poorly formed. The LoRA rank was also increased from r=16 to r=32, which doubles the number of trainable parameters from roughly 42 million to 84 million, representing about 1.15 percent of the model total weights.

Run	Training examples	ROUGE-L	BERTScore F1
First	380	0.1375	0.6814
Second	580	0.1612	0.7043

Table 2: Comparison of the two fine-tuning runs on 50 test questions

Both metrics improved consistently. The gain is modest but expected given the limited increase in training data. More data and a higher-capacity adapter produce answers that are lexically closer to the reference (ROUGE-L) and semantically more accurate (BERTScore).

Retrieval Component

The RAG pipeline retrieves the most relevant legal articles before passing them as context to the generation model. The quality of this retrieval step sets a hard ceiling on generation quality.

Baseline performance

The initial embedding model was evaluated on the 222 test questions. For each question, we measure how often the correct article appears among the top-k retrieved results.

Evaluation level	Baseline (mpnet)
Correct article in top 1	11.7% of questions
Correct article in top 3	20.3% of questions
Correct article in top 5	26.1% of questions
Correct article in top 10	34.7% of questions

Table 3: Baseline retrieval performance on 222 test questions

A Recall@10 of 34.7% means that for roughly two questions out of three, the correct article does not appear in the context provided to the model. This is the main bottleneck of the pipeline.

Upgraded embedding model

The embedding model was replaced with intfloat/multilingual-e5-large, a model that achieves better performance on multilingual retrieval benchmarks. It requires prepending the prefix `query:` to questions and `passage:` to documents before encoding. Re-indexing the full BSARD corpus with this model took approximately 45 minutes on the T4 GPU.

Evaluation level	mpnet	E5-large
Correct article in top 1	11.7%	19.8%
Correct article in top 3	20.3%	34.7%
Correct article in top 5	26.1%	42.3%
Correct article in top 10	34.7%	52.3%

Table 4: Retrieval performance before and after the embedding model upgrade

Recall@10 goes from 34.7% to 52.3%, a gain of nearly 18 percentage points. The correct article now appears in the top-10 results for more than half the test questions.

System Comparison

Three configurations were evaluated on the same 50 test questions to measure the contribution of each approach.

Configuration	ROUGE-L	BERTScore F1
Mistral base, zero-shot	0.0821	0.6112
RAG with upgraded retrieval	0.1535	0.6589
Fine-tuned Mistral (second run)	0.1612	0.7043

Table 5: Three-system comparison on 50 test questions

Mistral without any adaptation performs clearly worst, confirming that domain specialisation is necessary. RAG and fine-tuning independently reach similar ROUGE-L scores, which is consistent with findings in the literature such as Bouvard et al. (APIA 2024), who observed close answer relevance between the two approaches on French-language corpora. Fine-tuning shows a stronger advantage on BERTScore, suggesting the adapted model produces answers that are semantically closer to real legal articles.

Next Steps

The next priorities are running the human evaluation (blind comparison of RAG against fine-tuned responses), extending the error analysis to all three configurations, and starting the final report. The combination of both approaches into a single system is also planned for the coming weeks.